2025 MIT Science Bowl High School Invitational

Round 6

TOSS-UP

1) MATH – *Short Answer* Tim flips a fair coin 10 times. Given that the first 3 flips all landed on the same side, what is the expected number of total heads flipped?

ANSWER: 5

BONUS

1) MATH – *Short Answer* Tim attempts to find the sum of the first 100 positive integers with his calculator, but realizes afterwards that he accidentally added a 0 to the end of one number. Given the final sum was 5599, which number did he input incorrectly?

ANSWER: 61

TOSS-UP

2) ENERGY – *Short Answer* Researchers from the Sauer and Davis Labs studied a pathway for protein degradation in eukaryotic mitochondria. Most proteins in eukaryotes are tagged for degradation by covalently linking what small regulatory protein?

ANSWER: UBIQUITIN

BONUS

2) ENERGY – *Multiple Choice* Researchers at the Page lab have been studying sex differences in the heart rate of humans and other animals. Which of the following animals would have a heart most similar to that of a human?

W) Turtles
X) Birds
Y) Fish
Z) Squids

ANSWER: X) BIRDS

TOSS-UP

3) BIOLOGY – *Multiple Choice* A woman passes on the slay trait to all six of her children, yet her brother does not pass on the trait to any of his 54 offspring. What inheritance pattern is most likely?

W) X linked recessive
X) Autosomal recessive
Y) Mitochondrial
Z) Y linked

ANSWER: Y) MITOCHONDRIAL

BONUS

3) BIOLOGY – *Multiple Choice* Emilia is studying a certain eukaryote. During mitosis the spindles pass through the nuclear envelope through cytoplasmic channels, and the nucleus does not dissolve. Which of the following organisms could this be?

W) Dinoflagellate (read: *dino-flagellate*)
X) Yeast
Y) Diatom
Z) Fern

ANSWER: W) DINOFLAGELLATE (READ: *DINO-FLAGELLATE*)

TOSS-UP

4) PHYSICS – *Multiple Choice* A spring-mass system is placed on a frictionless inclined plane so that it oscillates parallel to the incline. As the inclination of the plane increases from 0 to 90 degrees, how does the period of oscillations of the spring-mass system change?

W) Uniformly increases
X) Uniformly decreases
Y) Increases, then decreases
Z) Remains constant

ANSWER: Z) REMAINS CONSTANT

BONUS

4) PHYSICS – *Short Answer* An infinite, uniformly charged sheet produces a constant electric field with strength 9 million volts per meter. A 1 kilogram particle carrying a 1 microcoulomb charge is released from rest 2 meters above the sheet. To the nearest meter per second, what is the particle's speed when it collides with the sheet?

ANSWER: 6

TOSS-UP

5) CHEMISTRY – *Multiple Choice* Which of the following group 14 oxides has the highest normal melting point?

W) Carbon dioxide
X) Silicon dioxide
Y) Germanium dioxide
Z) Tin dioxide

ANSWER: X) SILICON DIOXIDE

BONUS

5) CHEMISTRY – *Short Answer* Rank the following three organic compounds by increasing pKa: 1) Ethanol (read: *ETH-uh-noll*); 2) Acetaldehyde (read: *a-suh-TAL-duh-hide*); 3) Ethane (read: *ETH-ayn*).

ANSWER: 1, 2, 3

TOSS-UP

6) EARTH AND SPACE – *Multiple Choice* Which of the following correctly describes the role of latent heat in a developing mid-latitude cyclone?

W) Latent heat is absorbed during condensation, weakening the cyclone
X) Latent heat is released during condensation, strengthening the cyclone
Y) Latent heat is absorbed during precipitation, weakening the cyclone
Z) Latent heat is released during precipitation, strengthening the cyclone

ANSWER: X) LATENT HEAT IS RELEASED DURING CONDENSATION, STRENGTHENING THE CYCLONE

BONUS

6) EARTH AND SPACE – *Short Answer* Identify all of the following three depositional features that would typically be sorted: 1) Drumlin; 2) Floodplain; 3) Esker.

ANSWER: 2 AND 3

TOSS-UP

7) PHYSICS – *Short Answer* Two parallel plates are separated by a distance of 1 centimeter and carry opposite charges of 5 coulombs, generating a uniform electric field with magnitude 10 volts per meters between both plates. To two significant figures and in joules, how much work is required to separate the plates to an infinite distance?

ANSWER: 0.25

BONUS

7) PHYSICS – *Multiple Choice* A uniform cylinder 8 inches tall and 6 inches in diameter is placed vertically on an inclined plane. Assuming that the cylinder tips before it slips, which of the following is closest to the smallest angle in degrees from the horizontal needed for the cylinder to tip?

W) 30
X) 37
Y) 53
Z) 60

ANSWER: X) 37

TOSS-UP

8) EARTH AND SPACE – *Multiple Choice* Which of the following best describes the formation mechanism of a foreland basin?

W) Crustal flexure due to the weight of adjacent mountain belts
X) Crustal bending between a subduction zone trench and volcanic arc
Y) Thermal subsidence and sediment accumulation along a continental margin
Z) Downwarping of continental crust within a stable interior

ANSWER: W) CRUSTAL FLEXURE DUE TO THE WEIGHT OF ADJACENT MOUNTAIN BELTS

BONUS

8) EARTH AND SPACE – *Short Answer* Order the following three chemical species by increasing average concentration in Earth's oceans: 1) Carbonic acid; 2) Bicarbonate ion; 3) Carbonate ion.

ANSWER: 1, 3, 2

TOSS-UP

9) BIOLOGY – *Short Answer* In the beta oxidation of fatty acids, a certain two carbon intermediate is released upon each oxidation cycle. What is the identity of this two carbon intermediate?

ANSWER: ACETYL-COA

BONUS

9) BIOLOGY – *Short Answer* Identify all of the following three agents that are involved in the positive feedback cycle during labor: 1) Prostaglandins; 2) Oxytocin; 3) PYY.

ANSWER: 1 AND 2

TOSS-UP

10) ENERGY – *Short Answer* The Weiss Group recently reported a study on ocean detection in Triton. These oceans are predicted to be nitrogen-rich because of the presence of what molecule?

ANSWER: AMMONIA

BONUS

10) ENERGY – *Short Answer* Chemists at the Kulik Group studied spin-crossover between iron (II) and cobalt (II) metal centers. What is the spin of an iron (II) metal center that is octahedrally coordinated by strong-field ligands?

ANSWER: 0

TOSS-UP

11) CHEMISTRY – *Short Answer* At constant pressure, the density of an ideal gas is proportional to temperature raised to what power?

ANSWER: -1

BONUS

11) CHEMISTRY – *Short Answer* The reduction potential for the reduction of cerium (IV) to cerium (III) is +1.6 volts, while the reduction potential for the reduction of cerium (III) to metallic cerium is -2.4 volts. To one decimal place, what is the reduction potential for the reduction of cerium (IV) to metallic cerium in volts?

ANSWER: -1.4

TOSS-UP

12) MATH – *Multiple Choice* Hua is simplifying the expression $\sqrt[3]{60x}$ (read: *the cube root of the quantity sixty x*). For which of the following values of x will Hua not be able to simplify the expression by taking some factors out of the cube root?

W) 6
X) 9
Y) 12
Z) 15

ANSWER: Z) 15

BONUS

12) MATH – *Multiple Choice* A large n-by-n-by-n cube is painted maroon on all faces and cut into n^3 unit cubes. If a unit cube is selected uniformly at random, what is the expected number of maroon faces on the unit cube?

W) $1/6n^2$ (read: *1 divided by the quantity 6 times n squared*)
X) $6/n^2$ (read: *6 divided by the quantity n squared*)
Y) $1/6n$ (read: *1 divided by 6n*)
Z) 6/n (read: *6 divided by n*)

ANSWER: Z) 6/N (READ: *6 DIVIDED BY N*)

TOSS-UP

13) PHYSICS – *Multiple Choice* The planets Timorax and Timdra orbit the same star, with semi-major axes of length 6 astronomical units and 3 astronomical units, respectively. Which of the following is closest to the ratio of the periods of Timorax's orbit and Timdra's orbit?

W) 1
X) 1.4
Y) 2
Z) 2.8

ANSWER: Z) 2.8

BONUS

13) PHYSICS – *Short Answer* A conducting sphere is initially carrying a 54 microcoulomb charge. Charge begins to leak out of the sphere into the surrounding environment, which has constant resistivity. After one day, the sphere has a charge of 36 microcoulombs. After another day, what will the charge of the sphere be in microcoulombs?

ANSWER: 24

TOSS-UP

14) EARTH AND SPACE – *Multiple Choice* Which of the following pairs of objects do not possess stable Lagrange points associated with them?

W) Sun, Venus
X) Earth, Moon
Y) Saturn, Titan
Z) Pluto, Charon

ANSWER: Z) PLUTO, CHARON

BONUS

14) EARTH AND SPACE – *Short Answer* Order the following three spectral classes of stars by increasing strength of their Balmer lines: 1) G; 2) M; 3) A.

ANSWER: 2, 1, 3

TOSS-UP

15) CHEMISTRY – *Short Answer* How many distinct signals are in the carbon-13 NMR spectrum of hexamethylbenzene?

ANSWER: 2

BONUS

15) CHEMISTRY – *Short Answer* Rank the following three real gases by increasing *a* constant according to the Van der Waals equation: 1) Argon; 2) Chlorine; 3) Carbon dioxide.

ANSWER: 1, 3, 2

TOSS-UP

16) MATH – *Multiple Choice* The equations $y = e^x - 1$ and $y = \sqrt{x}$ are graphed on the x-y plane. Which of the following is the number of intersection points between these two graphs?

W) 0
X) 1
Y) 2
Z) 3

ANSWER: Y) 2

BONUS

16) MATH – *Short Answer* 1000 robots are placed in a room, 500 of which are currently humming. Each second, any robot that is humming has a 15% chance to stop humming, and any robot that is quiet has a 10% chance to begin humming. After a long time, how many robots are humming in expectation?

ANSWER: 400

TOSS-UP

17) ENERGY – *Short Answer* Scientists from the Earthquake Science Group used deep learning to detect unrecorded low-frequency earthquakes. While training a deep neural network, what technique based on the chain rule is used to iteratively compute the gradients of the loss function with respect to each layer in the network?

ANSWER: BACKPROPAGATION

BONUS

17) ENERGY – *Short Answer* Researchers in the DeCoDe Lab at MIT are studying machine learning models. What type of model is trained on vast datasets so it can be applied in a large range of use cases?

ANSWER: FOUNDATION MODEL (ACCEPT: PFN)

TOSS-UP

18) BIOLOGY – *Short Answer* While in mammals, females have separate external openings for the reproductive, digestive, and excretory systems, nonmammalian vertebrates often have a common opening for these three tracts. What is the name of this structure, where eggs exit chickens?

ANSWER: CLOACA

BONUS

18) BIOLOGY – *Short Answer* In the breakdown of erythrocytes, the iron-porphyrin complex is broken down into Fe2+ and what pigmented compound that is a component of bile?

ANSWER: BILIRUBIN

TOSS-UP

19) PHYSICS – *Short Answer* A 1 kilogram point mass is attached to a 1 meter rod to form a pendulum, which is initially inverted with the mass directly above the pivot. The mass is nudged slightly and eventually comes to rest directly below the pivot due to air resistance. To the nearest tenth and in joules, what is the total work done by air resistance on the point mass?

ANSWER: -19.6

BONUS

19) PHYSICS – *Short Answer* A ball is launched with initial horizontal velocity 7 meters per second and vertical velocity 8 meters per second. In meters per second, what is the speed of the ball when it reaches half of its maximum height?

ANSWER: 9

TOSS-UP

20) EARTH AND SPACE – *Multiple Choice* Which of the following types of variable stars is not driven by the kappa mechanism?

W) RR Lyrae (read: *LYE-ree*)
X) Cepheid (read: *SEE-fee-id*)
Y) Mira (read: *MEE-rah*)
Z) Delta Scuti (read: *SKOO-tee*)

ANSWER: Y) MIRA (READ: *MEE-RAH*)

BONUS

20) EARTH AND SPACE – *Short Answer* Mercury completes one rotation about its axis in 58.7 Earth days. To the nearest whole number, how many Earth days does Mercury take to complete one revolution about the Sun?

ANSWER: 88

TOSS-UP

21) ENERGY – *Multiple Choice* The Johnson Group is studying triboluminescence in photonic crystals. Which of the following causes light to be produced in triboluminescence?

W) Thermal excitation
X) Sound waves
Y) Magnetic fields
Z) Mechanical stress

ANSWER: Z) MECHANICAL STRESS

BONUS

21) ENERGY – *Short Answer* Physicists at the Quantum Measurement Group are using spectroscopy to probe quantized vibrational modes in molecular solids. What quasiparticle are these vibrational modes?

ANSWER: PHONONS

TOSS-UP

22) CHEMISTRY – *Multiple Choice* Under the Born-Oppenheimer approximation, which of the following terms in the Hamiltonian for a neutral helium atom makes finding an exact solution to the Schrodinger equation an intractable problem?

W) Kinetic energy of electrons
X) Kinetic energy of nucleus
Y) Attraction between nucleus and electrons
Z) Repulsion between electrons

ANSWER: Z) REPULSION BETWEEN ELECTRONS

BONUS

22) CHEMISTRY – *Short Answer* Copper (II) hydroxide is very poorly soluble in water. Identify all of the following three 1 molar aqueous solutions that copper (II) hydroxide would be more soluble in than pure water: 1) Hydrochloric acid; 2) Sodium hydroxide; 3) Ammonia.

ANSWER: 1 AND 3

TOSS-UP

23) BIOLOGY – *Short Answer* Order the following three enzymes in chronological order encountered in the digestive tract: 1) Amylase; 2) Trypsin; 3) Pepsin.

ANSWER: 1, 3, 2

BONUS

23) BIOLOGY – *Short Answer* Hector finds an animal zygote and during the 4 cell stage removes one of the 4 cells. The resulting cells develop into an adult animal that is missing major organs. Identify all of the following three phyla that this animal could belong to: 1) Echinodermata; 2) Chordata; 3) Nematoda.

ANSWER: 3 ONLY

TOSS-UP

24) MATH – *Short Answer* Adeline says that "the absolute value of the secant of theta is greater than a if and only if the absolute value of the cosecant of theta is less than a." What is the unique real value of a for which Adeline's statement is always true?

ANSWER: $\sqrt{2}$ (READ: *THE SQUARE ROOT OF TWO*)

BONUS

24) MATH – *Short Answer* What is the tens digit of 11^{2025} (read: *11 raised to the power of 2025*)?

ANSWER: 5

TOSS-UP

25) PHYSICS – *Short Answer* A particle in one dimension moves in the potential $U(x) = x^3 - 3x$ (read: *U of x equals x cubed minus three x*). How many equilibrium points does this system have?

ANSWER: 2

BONUS

25) PHYSICS – *Short Answer* Identify all of the following three materials that have an excess of electron holes compared to electrons: 1) Pure silicon; 2) Silicon doped by trace nitrogen; 3) Silicon doped by trace boron.

ANSWER: 3 ONLY